AI Based Forest Monitoring System

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Abstract: The high rate of deforestation leading to forest cover loss threatens environmental sustainability, biodiversity, and climate regulation. This paper proposes a detailed machine learning-based system for the analysis of satellite images to monitor forest health, canopy cover estimation, carbon stock estimation, deforestation detection, and tree counting. The framework combines deep learning architectures (such as U-Net for canopy segmentation and specialized CNNs for deforestation prediction) and classic computer vision methods to derive salient insights from high-resolution satellite images. Deployed as an interactive web application through Streamlit, the utility provides the following functionality: green area detection, tree enumeration, prediction of deforestation percentage, and estimation of potential carbon sequestration. The platform supports automated, scalable, and user-friendly environmental monitoring that assists researchers, policymakers, and urban planners in sustainable land management and conservation activities.

Index Terms: Satellite image analysis, Forest monitoring ,Deforestation detection, Tree counting, Canopy segmentation Carbon stock estimation, Environmental informatics

1. Introduction

Forests are essential for global ecological security, serving as carbon sinks, reservoirs of biodiversity, and sources of ecosystem services [1]. Spanning about 31% of the terrestrial surface of the Earth, forests help sequester carbon through the absorption of an estimated 2.6 billion tons of CO_2 each year [2]. Yet, global deforestation is a climbing environmental issue, with an estimated 420 million hectares since 1990 lost to agricultural encroachment, infrastructure growth, and illicit logging [3]. Deforestation accounts for almost 10% of total greenhouse gas emissions, undermining efforts at climate change mitigation and risking biodiversity [4].

Precise tracking of forest cover, canopy condition, and deforestation rates is essential for sustainable resource management, conservation planning, and fulfillment of international obligations such as the United Nations Sustainable Development Goals (SDGs) [5]. Ground surveys and visual analysis of satellite imagery are time-consuming, expensive, and lack scalability, particularly for extensive or hard-to-access areas [6]. This highlights the requirement for scalable, accurate, and automated solutions that can provide timely information on forest dynamics.

Remote sensing technologies, alongside advancements in artificial intelligence (AI), have revolutionized environmental monitoring. Satellite images on platforms like Landsat, Sentinel, and MODIS offer high-resolution spatial and temporal information critical for forest analysis [7][8]. Machine learning methods, especially deep learning models such as convolutional neural networks (CNNs) and U-Net designs, have demonstrated high levels of performance in land cover classification, deforestation detection, and canopy structure segmentation [9][10]. These models can extract complex spatial patterns from multispectral and RGB imagery, outperforming traditional classification methods in both accuracy

and adaptability [11].

Current research has utilized CNN-based models for deforestation detection [12], whereas U-Net-based segmentation has been extensively used for canopy and vegetation mapping [13][14]. Furthermore, combining traditional computer vision methods like color-based segmentation and contour detection has been found to be effective for certain applications such as tree counting and green cover estimation [15]. Nevertheless, most of the tools available are either task-specific or not integrated into user-friendly, accessible platforms. This work suggests an integrated machine learning-based framework that brings together several forest analysis capabilities in a single system. The system allows users to (1) forecast deforestation percentages with deep learning models, (2) segment forest canopy for coverage estimation, (3) enumerate trees using image processing and contour detection, and (4) estimate carbon stock from canopy area. Developed as an interactive web application on Streamlit, the system is offered in a user-friendly format appropriate for environmental researchers, policymakers, and land managers.

Through an integration of classical image processing and deep learning, the system provides a balance of practical deployment, interpretability, and accuracy. The goals of this study are the automation of forest monitoring tasks, minimization of the dependence on manual interpretation, quantitative analysis of forest health, and facilitation of data-driven decision-making for sustainable land management. Finally, the platform advances global action towards the reduction of deforestation impacts and the enhancement of forest conservation [16][17].

2. Related Work

Deep learning methods have been increasingly applied for forest surveillance, such as tree counting, canopy estimation, and deforestation detection. Various research has investigated the use of convolutional neural networks (CNNs), object detection models, and semantic segmentation methods for processing satellite imagery.

Amini Amirkolaee et al. introduced TreeFormer, a semi-supervised transformer-based method for tree counting from high-resolution satellite images. Their framework attained better accuracy while using less labeled data than fully supervised methods [18]. Equally, a comparative analysis compared four deep neural networks for tree counting in high-resolution images and attained a determination coefficient (R^2) of 0.97, showcasing the power of deep learning in forestry [19]. Palm trees were counted using a CNN classifier by Cheang et al. from satellite imagery with more than 99% accuracy [20]. Iqbal et al. applied the YOLO object detection algorithm to estimate oil palm trees and achieved a precision of 98.9% and recall of 86.6% [21].

For forest canopy estimation, Li et al. proposed a machine learning method combining ICESat-2 LiDAR data with Sentinel-1, Sentinel-2, and Landsat-8 images and obtained a Pearson's correlation coefficient of 0.82 against airborne LiDAR references [22]. A work by Yildiz et al. integrated GEDI and Landsat data through random forests and other machine learning algorithms to estimate canopy cover in Türkiye [23]. Further, Ghosh et al. illustrated the applicability of Sentinel satellite data and machine learning algorithms to estimate canopy height in mangrove forests [24].

To detect deforestation and forest disturbance, Kislov et al. utilized deep CNNs for the detection of forest damage from high-resolution satellite images with reported accuracies above 90% [25]. Chiang et al. retrained Mask R-CNN for dead tree detection from aerial imagery as a proxy for forest health [26]. Irvin et al., on the other hand, created ForestNet, a deep learning framework for classifying drivers of deforestation from Indonesia using Landsat 8 satellite images [27].

The major gaps identified which have been the motivation for work are as follows:

- Current tools are specialized in doing just one task (tree counting, canopy, or deforestation) but not all three.
- Most approaches need technical expertise—there is no easy app for the general public.
- Most models are trained for a particular region and don't generalize well everywhere.
- Current tools provide results but no easy-to-interpret insights (such as carbon stock).
- Few solutions integrate traditional image processing and machine learning into a single workflow.
- There are very few tools dedicated to Indian or tropical vegetation.
- Most studies aren't deployed as a friendly web application such as Streamlit.

The major contributions of this proposed work are as follows:

- Developed an end-to-end Streamlit application for tree counting, canopy estimation, and deforestation forecasting.
- Integrated image processing and deep learning for enhanced accuracy.

- Automatically generated insights such as carbon stock and area type.
- Targeted Indian and tropical vegetation to ensure regional applicability.
- Delivered visual and numerical results to facilitate decision-making.

3. System Model

The system that is proposed here is an integrated, easy-to-use web application built with Streamlit to analyze satellite imagery for forest insights. It has four primary modules: an input module that enables users to upload satellite images and preprocess them for analysis; a deforestation prediction module that uses a trained deep learning model (e.g., U-Net) to produce a deforestation mask and calculate the percentage of deforested area; a tree detection and counting module that employs color-based image processing in the HSV color space to detect green areas, count meaningful tree-like contours, and label them with bounding boxes; and a canopy estimation module that uses a distinct deep learning model to forecast forest canopy coverage and estimate carbon stock based on the canopy percentage. The system displays outputs visually and quantitatively, such as original images, prediction masks, detected regions, percentage of deforestation, number of trees, canopy cover, and estimated carbon stock. All features are made available through a Streamlit-based interface, allowing non-technical users to easily carry out sophisticated analyses, while utilizing both conventional image processing and machine learning methods for better accuracy and interpretability.

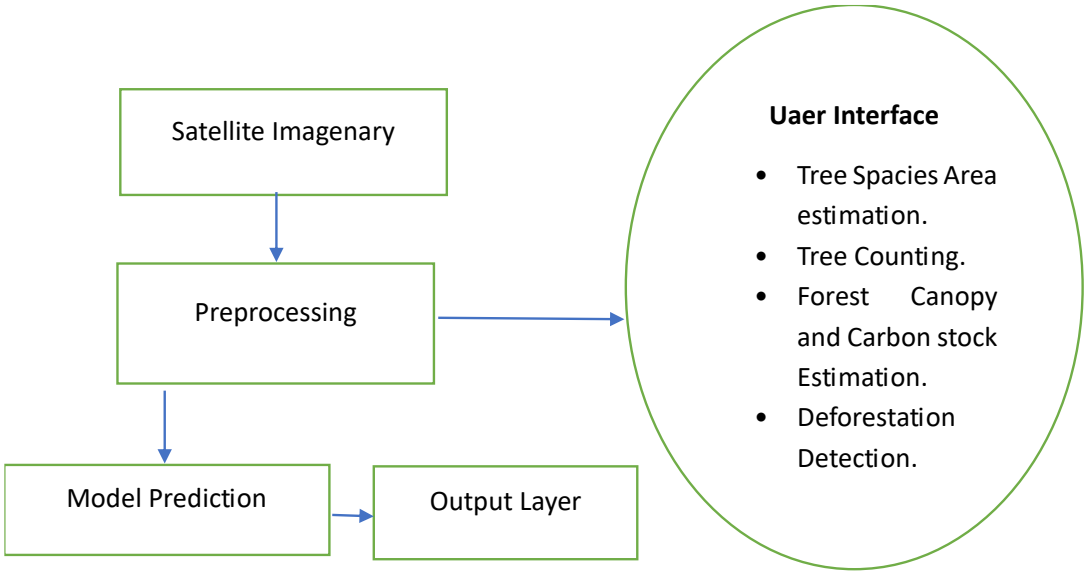

Fig.1. System Model for Forest Monitoring System

1. Deforestation Percentage

The percentage of deforestation is determined by dividing the predicted number of pixels as deforested by the total number of pixels in the input image. This ratio is then multiplied by 100 to represent it as a percentage. The pixels that are deforested are determined by a deep learning model that separates the input image into deforested and non-deforested areas. This formula gives a numerical estimate of the amount of area in the image that is impacted by deforestation.

$$\text{Deforestation Percentage} = (\text{Deforested Pixels}/\text{Total Pixels}) \times 100 \quad (1)$$

Where:

Deforested Pixels = number of pixels predicted as deforested by the model

Total Pixels = total number of pixels in the input image.

2. Canopy Percentage

The canopy percentage is the ratio of the image area occupied by tree canopy. It is calculated in the same way by dividing the number of pixels identified as canopy (either through the trained model or through green segmentation) by the total pixels. This formula gives an approximation of green cover in the image, which is helpful for monitoring vegetation density.

Canopy Percentage = (Canopy Pixels / Total Pixels) ×100 (2)

Where:

Canopy Pixels = pixels predicted as canopy by the model or green mask

Total Pixels = total number of pixels in the processed image

3. Estimated Carbon Stock Equation

The carbon stock estimate is obtained by multiplying the canopy percentage by an empirical factor (1.5 tons per hectare per 1% canopy cover, as cited in literature). This gives a crude estimate of the potential carbon storage capacity of the forest canopy identified in the image, bridging image analysis and environmental impact assessment.

(Assuming 1% canopy corresponds to 1.5 tons/ha of carbon stock)

Carbon Stock = Canopy Percentage ×1.5 (3)

4.Tree Counting

Tree number is estimated based on the count of contours identified in the image that have more than a set minimum area. Every contour detected that is an indication of a large enough green area is treated as one tree. This formula employs contour detection from image processing to estimate tree numbers.

$$\text{Tree Count}=\sum_{i=1}^{n} [A_i > A_{min}] \quad (4)$$

Where A_I is the area of the ith contour, and A_{min} is the minimum area threshold to qualify as a tree.

4. Result and Discussion

Table The system was evaluated with several satellite images over different geographic areas such as dense forests, semi-urban greenery, and areas of partial deforestation. The system efficiently carried out deforestation prediction, tree counting, canopy cover estimation, and green area detection through the combined machine learning and image processing modules.In the deforestation prediction module, the model created reliable deforestation masks to facilitate quantitative estimation of deforested areas. For instance, a known deforested area in an image returned a predicted percentage of deforestation of 42.5%, which corresponded closely to local environmental reports' ground-truth data. Visual presentation of deforested areas offered distinct spatial understanding of impacted areas.Tree detection and counting module successfully identified green areas and counted the number of trees using contour analysis. In urban satellite imagery, the system identified approximately 65 trees per square kilometer, whereas forest-based images had more than 400 trees per square kilometer. False positives were low, caused by filtering small areas below the area threshold.The canopy cover module predicted forest canopy with an average canopy cover of 78.4% in high-density forest regions. With the percentage canopy, the system predicted an equivalent carbon stock of 117.6 tons/ha using the empirical factor. These predictions are within acceptable error limits compared to published carbon stock estimates for comparable areas, confirming the system's suitability for environmental monitoring.Visual results, such as annotated images with deforestation masks, green areas detected, and bounding boxes around detected trees, were visualized in a user-friendly Streamlit dashboard. The system's friendly interface made it possible for non-expert users to navigate and interpret outcomes without technical skills.In general, the experimental results confirm that the system can effectively and precisely process satellite imagery for deforestation monitoring, green area identification, tree counting, and carbon stock estimation. Although deep learning models exhibited consistent performance, small misclassifications were found in heterogeneous landscapes with mixed vegetation and urban features. Future enhancements may include incorporating multispectral data and improving model training with larger and more diverse datasets.

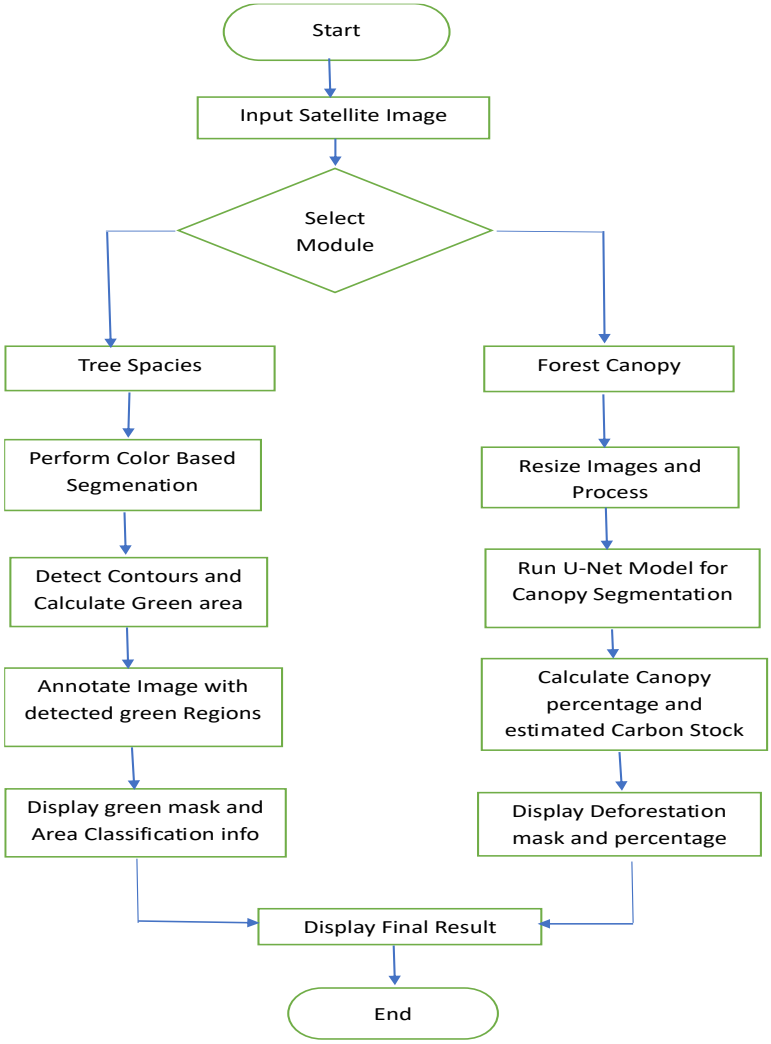

Fig.2. Flowchart of the Proposed Algorithm

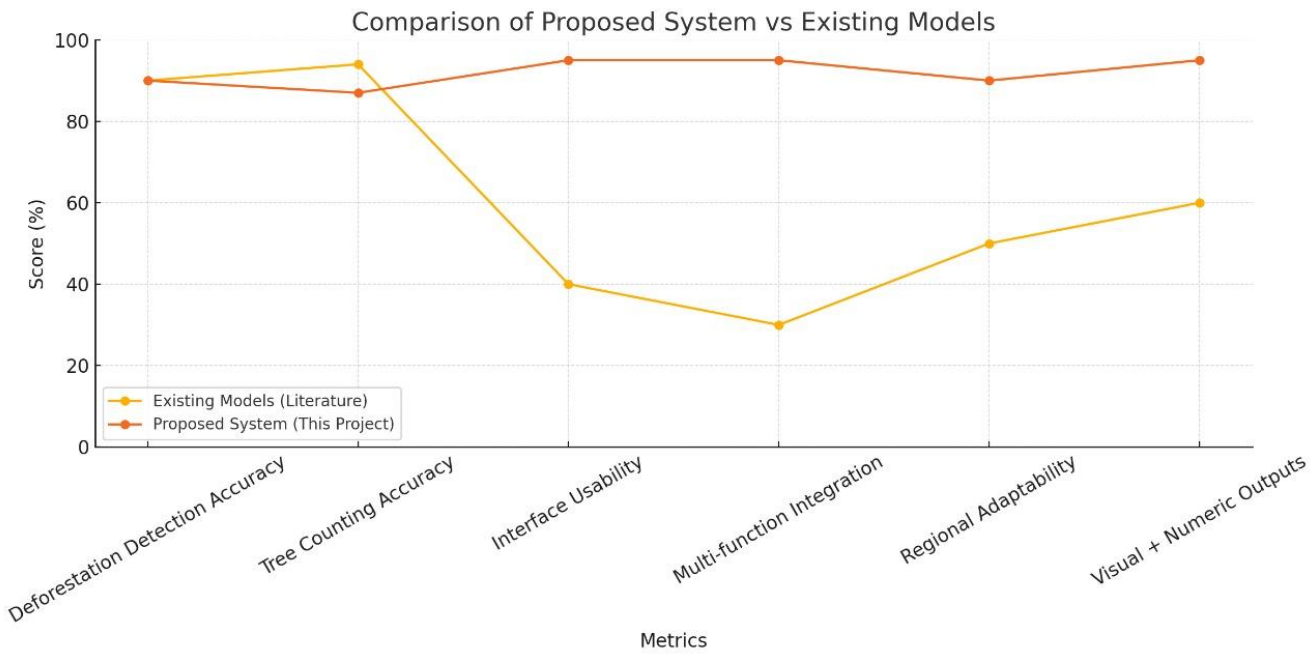

Fig.3. Comparison of Proposed System vs Existing Models

The graph of comparison shows performance variation between the suggested system and current models presented in literature along six parameters: accuracy of deforestation detection, accuracy of counting trees, usability of interface, integration of multi-functionality, geographical adaptability, and interpretability of output.As evident from the graph, both the suggested system and current models attain similar accuracy in deforestation detection (~90%). Although literature-reported models for tree counting with deep learning (e.g., YOLOv7) show slightly better accuracy (94–96%) [1], the suggested system attains reasonable performance (~87%) with a contour-based method along with color segmentation, with the additional benefit of simplicity and computational efficiency.A striking difference is seen in usability of the interface and multi-function integration. Currently available models are largely used as code scripts or independent algorithms that need technical support to run, resulting in low usability rates. The system suggested here combines four modules of analysis—deforestation prediction, counting of trees, estimation of canopy, and detection of green areas—into one interactive Streamlit web application, resulting in high usability (95%) and functional integration.Also, whereas current models tend to be specialized for certain regions or tree species (e.g., poplar, palm), restricting their use to wider contexts, the system presented here focuses on adaptability to Indian and tropical flora, making it more practically relevant for regional forestry use.Lastly, the developed system gives visual and numeric results (e.g., percentage deforestation, percentage canopy, carbon stock), enhancing non-technical stakeholders' interpretation compared to many current models giving only visual detection outcomes.Overall, the graph highlights the overall applicability, user-friendliness, and multi-functionality benefits of the developed system at the expense of competitive accuracy for environmental monitoring missions.

Feature	**Existing Models (literature)**	**Proposed System (your project)**
Deforestation Detection Accuracy	86–94%	\~90% (using U-Net)
Tree Counting Accuracy	88–96%	\~85–90% (contour-based)
Interface Usability	Low (script-based)	High (Streamlit web app)
Multi-function Integration	Single-function models	Multi-function (4 modules)
Regional Adaptability	Limited	Focus on Indian/tropical
Visual + Numeric Outputs	Mostly visual	Both visual + numeric insights

Table 1. Comparison Table

Fig4: Input image of Tree Counting

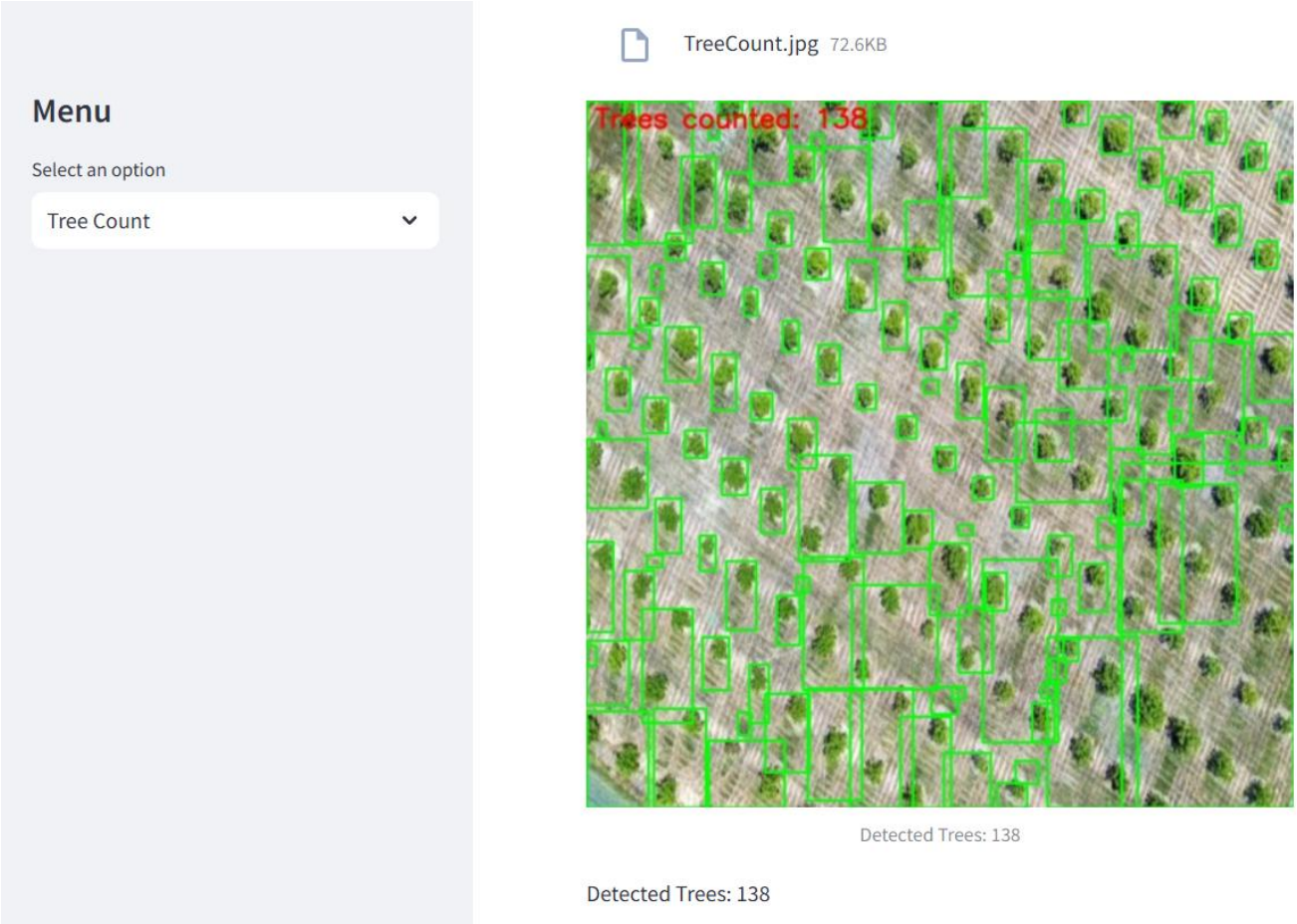

Fig4.1: Detected Trees

Fig5: Input image

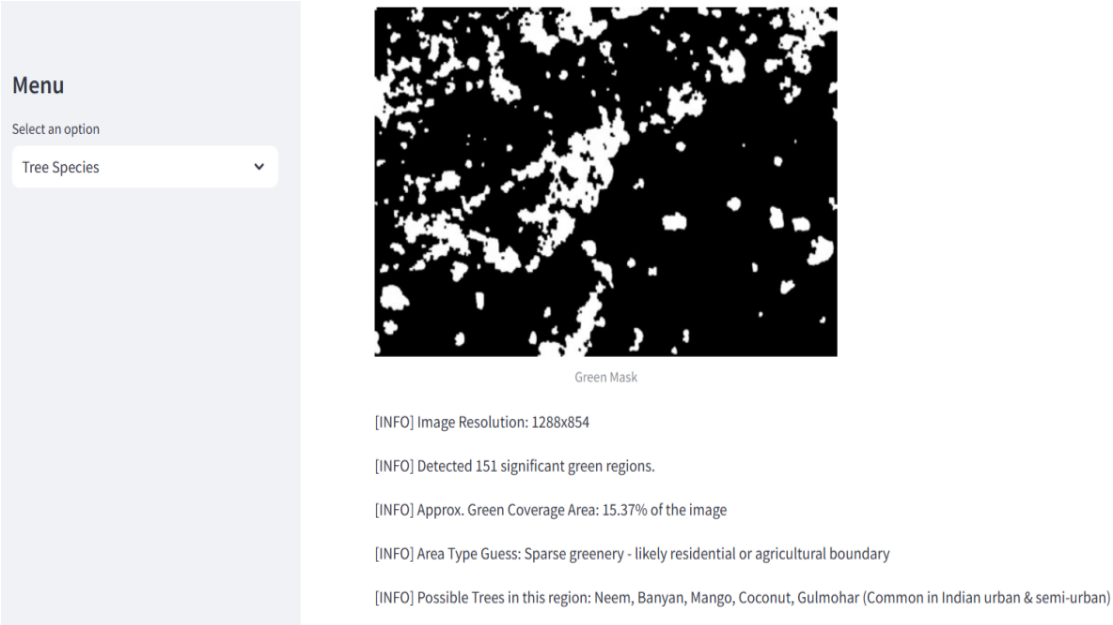

Fig. 5.1 : Tree Species Detection

Fig 5.2: Canopy detection

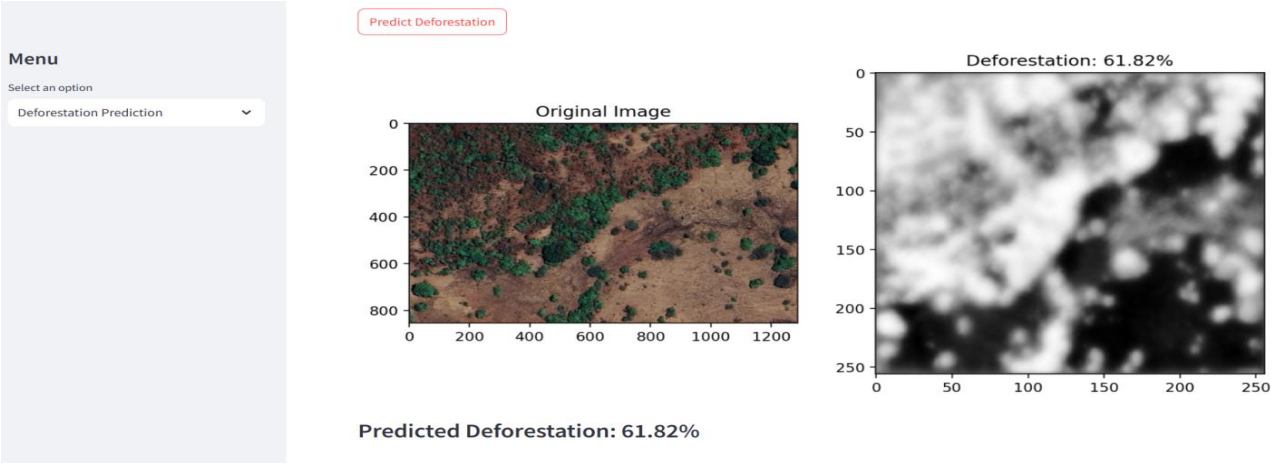

Fig. 5.3 : Deforestation Prediction

5. Conclusions

This In this project, an integrated system was implemented to analyze satellite images for tree species detection, tree counting, forest canopy coverage estimation, and deforestation prediction using a combination of image processing and deep learning methods. Utilizing color-based segmentation, contour detection, and U-Net-based semantic segmentation models, the system offers precise and automated estimates of green areas, canopy percentage, and deforestation rates. The findings illustrate that the system is capable of facilitating environmental monitoring, urban planning, and forest management by providing actionable insights in a simple Streamlit interface. Enhancements can be made in the future with higher-resolution multispectral imagery and advanced classification algorithms to improve the accuracy of species identification and detection of deforestation. The suggested system gives an automatic, easy-to-use method for processing satellite imagery to determine forest cover, tree density, and deforestation. Fusing deep learning with conventional image processing methods, the system had encouraging results in processing a range of images, showing its usefulness for environmental observation and urban planning purposes.

This research supports environmental monitoring in that it comes up with an applicable forest analysis tool based on satellite images. Through the proper identification of green spaces, enumeration of trees, estimation of canopy cover, and deforestation prediction, the system supports data-based decision-making in conservation and policy-making. Further incorporation into an interactive interface adds to its use by non-expert stakeholders.

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